

Integration WS 8

1. Evaluate

a. $\int \sqrt{x^2 + 1} dx$

b. $\int \sqrt{4 - x^2} dx$

2. Evaluate the following definite integrals:

a. $\int_0^1 x e^x dx$

b. $\int_0^{\frac{\pi}{2}} x \cos x dx$

c. $\int_1^e x \ln x dx$

d. $\int_0^{\frac{1}{2}} \sin^{-1} 2x dx$

e. $\int_0^{\frac{\pi}{2}} e^x \cos x dx$

f. $\int_0^{\frac{\pi}{2}} x \sin^2 x dx$

3. a. Use the substitution $y = \frac{2}{3} \sin \theta$ to find

$$\int_0^{\frac{2}{3}} \sqrt{4 - 9y^2} dy$$

b. Find $\int_0^1 \frac{\sin^{-1} y}{\sqrt{1+y}} dy$

4. a. Use integration by parts to show that

$$\int \cos^{-1} \left(\frac{x}{6} \right) dx = x \cos^{-1} \left(\frac{x}{6} \right) - 6 \sqrt{1 - \frac{x^2}{36}} + C$$

b. Show that $\int_0^1 \tan^{-1} u du = \frac{\pi}{4} - \frac{1}{2} \ln 2$

5. Evaluate:

a. $\int_0^{\frac{\pi}{3}} x \sec^2 x dx$

b. $\int_1^e \log_{10} x dx$

c. $\int_0^{\pi} x \cos x dx$

d. $\int_1^e \ln x dx$

e. $\int_e^{e^2} x \ln x dx$

f. $\int_e^{e^4} \frac{dx}{x \ln x}$

g. $\int_0^{\frac{\pi}{4}} x \sin 2x dx$

h. $\int_1^e \frac{\ln x}{x^4} dx$

i. $\int_0^1 t e^{-2t} dt$

j. $\int_0^1 u \tan^{-1} u du$

k. $\int_0^{\ln 2} t e^{-t} dt$

l. $\int_0^1 \sin^{-1} t dt$

m. $\int_0^{\frac{\pi}{4}} x \cos^2 x dx$

6. a. Determine $\int_{2+\epsilon}^4 \frac{dx}{x^2 \sqrt{x^2 - 4}}$, where $\epsilon > 0$

b. Take the limit of this result as $\epsilon \rightarrow 0^+$, and

hence find $\int_2^4 \frac{dx}{x^2 \sqrt{x^2 - 4}}$.

7. Find

a. $\int \frac{2x+3}{(x-1)(x-2)(2x-3)} dx$

b. $\int \frac{4x+12}{x^3 - 6x^2 + 8x} dx$

8. Evaluate

a. $\int_2^7 \frac{3x+5}{(x-1)(x+2)(x+1)} dx$

b. $\int_1^2 \frac{13x+6}{x^3 - x^2 - 6x} dx$

9. a. i. Let $\frac{2x^2+1}{(x-1)(x+2)} = A + \frac{B}{x-1} + \frac{C}{x+2}$. Find the values of A, B and C .

ii. Hence find $\int \frac{2x^2+1}{(x-1)(x+2)} dx$

b. Use a similar technique to part (a) in order to find:

i. $\int \frac{x^2 - 2x + 3}{(x+1)(x-2)} dx$

ii. $\int \frac{3x^2 - 66}{(x+4)(x-5)} dx$

10. a. i. Find the integer A, B, C and D such that

$$\frac{x^3 - 3x^2 - 4}{(x+1)(x-3)} = Ax + B + \frac{C}{x+1} + \frac{D}{x-3}$$

ii. Hence evaluate $\int_0^1 \frac{x^3 - 3x^2 - 4}{(x+1)(x-3)} dx$

b. Use a similar method to evaluate

$$\int_2^4 \frac{x^3 + 4x^2 + x - 3}{(x+2)(x-1)} dx$$

11. Show that

$$\int_0^{\pi} \left(x - \frac{\pi}{2} \right)^6 \cos^3 x dx = - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^6 \sin^3 x dx = 0$$

12. $I_n = \int_0^{2\pi} (1 + \cos \theta)^n d\theta$ for $n \geq 0$

- a. Show that $I_{n+1} = \frac{2n+1}{n+1} I_n$
b. Find I_3
13. Given $I_n = \int_0^1 \frac{x^n}{\sqrt{x+1}} dx$ for $n = 0, 1, 2, 3 \dots$
a. Show that $(2n+1)I_n = 2\sqrt{2} - 2nI_{n-1}$ for $n = 1, 2, 3 \dots$
14. Evaluate $\int_0^1 \frac{x^3}{\sqrt{x+1}} dx$ $I_n = \int_1^e (1 - \ln x)^n dx$ for $n = 0, 1, 2 \dots$
a. Show that $I_n = -1 + nI_{n-1}$ for $n = 1, 2, 3 \dots$
b. Hence evaluate $\int_1^e (1 - \ln x)^3 dx$
c. Show that $\frac{I_n}{n!} = e - \sum_{r=0}^n \frac{1}{r!}$ For $n = 1, 2, 3 \dots$
d. Show that $0 \leq I_n \leq e - 1$
e. Deduce that $\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{r!} = e$
15. For $n = 0, 1, 2 \dots$ let $I_n = \int_{e^{-1}}^1 (1 + \ln x)^n dx$ and $J_n = \int_{e^{-1}}^1 \ln x (1 + \ln x)^n dx$
a. Show that $I_n = 1 - nI_{n-1}$ for $n = 1, 2, 3 \dots$
b. Show that $J_n = 1 - (n+2)I_n$ for $n = 0, 1, 2 \dots$
c. Hence find the value of J_3 is simplest exact form
16. Using an appropriate substitution, prove that $\int_0^{2a} f(x) dx = \int_{-a}^a f(a-x) dx$ for $a > 0$
a. Hence show $\int_0^1 \sqrt{x(1-x)} dx = \frac{\pi}{8}$
b. If $I_n = \int_0^1 \sqrt{x(1-x)}^{\frac{n}{2}} dx$, $n = 0, 1, 2 \dots$ show that $I_n = \frac{n}{n+3} I_{n-2}$ for $n \geq 2$
c. Hence evaluate $\int_0^1 \sqrt{x(1-x)}^{\frac{5}{2}} dx$
17. a. Find $\int \frac{x^2}{(x-1)(x-2)} dx$
b. Find:
i. $\int \cos^3 x dx$
ii. $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$
c. Show that $\int_0^1 \frac{\sqrt{x}}{1+x} dx = 2 - \frac{\pi}{2}$, and hence evaluate $\int_0^1 \frac{1}{\sqrt{x}} \ln(1+x) dx$
18. a. Find $\int \frac{4}{x^2+2x-1} dx$
b. If $a > 0$, use the substitution $u = \frac{1}{x}$ to evaluate $\int_{\frac{1}{a}}^a \frac{\ln x}{x^2+1} dx$
c. If $I_n = \int_0^1 (1-x^2)^n dx$ for $n \geq 0$, show that $I_n = \frac{2n}{2n+1} I_{n-1}$ for $n \geq 1$. Hence, or otherwise, show that $I_n = \frac{2^{2n}(n!)^2}{(2n+1)!}$ For $n \geq 1$.

Answer

1. a. $\frac{1}{2} [(x\sqrt{x^2+1} + \ln(x + \sqrt{x^2+1}))]$; b. $\frac{1}{2} x\sqrt{4-x^2} + 2 \sin^{-1} \frac{x}{2}$
2. a. 1; b. $\frac{\pi}{2} - 1$; c. $\frac{e^2}{4} + \frac{1}{4}$; d. $\frac{\pi}{4} - \frac{1}{2}$; e. $\frac{1}{2} (e^{\frac{\pi}{2}} - 1)$; f. $\frac{\pi^2+4}{16}$
3. a. $\frac{\pi}{3}$; b. $\sqrt{2}\pi - 4$
5. a. $\frac{\pi}{\sqrt{3}} - \ln 2$; b. $\frac{1}{\ln 10}$ or $\ln_{10} e$; c. -2;
d. 1; e. $\frac{e^2}{4} (3e^2 - 1)$; f. $\ln 4$; g. $\frac{1}{4}$; h. $\frac{1-4e^{-3}}{9}$; i. $-\frac{3}{4} e^{-2} + \frac{1}{4}$; j. $\frac{\pi}{4} - \frac{1}{2}$; k. $\frac{1}{2} (1 - \ln 2)$; l. $\frac{\pi}{2} - 1$; m. $\frac{\pi^2+4\pi-8}{64}$
6. a. $\frac{\sqrt{3}}{8} - \frac{\sqrt{\epsilon(4+\epsilon)}}{4(2+\epsilon)}$; b. $\frac{\sqrt{3}}{8}$
7. a. $5 \ln(x-1) + 7 \ln(x-2) - 12 \ln(2x-3) + C$; b. $\frac{3}{2} \ln x - 5 \ln(x-2) + \frac{7}{2} \ln(x-4) + C$
8. a. $\frac{5}{3} \ln 3 - \ln 2$; b. $2 \ln 3 - 8 \ln 2$
9. a. i. $A = 2, B = 1, C = -3$, ii. $2x + \ln(x-1) - 3 \ln(x+2) + C$;
b. i. $x + \ln(x-2) - 2 \ln(x+1) + C$, ii. $3x + 2 \ln(x+4) + \ln(x-5) + C$
10. a. i. $A = 1, B = -1, C = 2, D = -1$,
ii. $\ln 3 + \ln 2 - \frac{1}{2}$; b. $12 + \ln 2$
12. $I_3 = 5\pi$
13. $\frac{32}{105} - \frac{18\sqrt{2}}{105}$
14. $\int_1^e (1 - \ln x)^3 dx = -16 + 6e$
15. $J_3 = 11 - \frac{30}{e}$
16. $I_5 = \frac{5\pi}{128}$
17. a. $x + \ln \left| \frac{(x-2)^4}{x-1} \right|$; b. $\sin x - \frac{1}{3} \sin^3 x, \frac{1}{2} (\sin^{-1} x)^2$; c. $2 \ln 2 - 4 + \pi$
18. a. $\sqrt{2} \ln \left| \frac{x+1-\sqrt{2}}{x+1+\sqrt{2}} \right|$; b. 0